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Problem Set 2

I. FORWARD PRICES OF BONDS (AND THE FORWARD CURVE)

Recall that the yield curve gives the rate to borrow at different maturities. So if r_T is the (annual) T-year yield, then the price today ($t = 0$) of a bond[1] that pays off T years from now (at $t = T$) is

$$p_0 = \frac{1}{(1 + r_T)^T}.$$

You can find the US yield curve here:

www.treasury.gov/resource-center/data-chart-center/interest-rates/

1. Given the yield curve, what is the price of:
   a) A 1-year bond?

   b) A 10-year bond?

[1] Remember that here a "bond" is a zero-coupon bond with face value 1.

2. Recall the forward price $F_0(T) = (1 + r_T)^T p_0$, where r_T is the annualized T-year yield.

   Today, what is the one-year forward price of:

   a) The 1-year bond?

   b) The 10-year bond?

II. Synthetic Risk-free Rate

Suppose cocoa is trading in the market at \$1. The one-year forward price is \$1.70.

1. Roughly sketch the payoff diagrams of the forward, the underlying, and the bond (at maturity).

2. Replicate the payoff of the bond with a portfolio of the forward and cocoa. Illustrate the replication graphically.

   Note that you have created a "synthetic" bond, i.e. you can use this portfolio to borrow and lend at the risk-free rate without trading in the bond directly.

3. Suppose there is no storage cost/convenience yield. Using your answer above, find the 1-year risk-free rate.

4. Compare the rate you found above to the actual 1-year Treasury rate. Are your answers the same? If you could trade in actual Treasuries, could you construct an arbitrage? If so, how? (You don't have to show it is an arbitrage, just say the strategy.)

III. FORWARD PRICING

There are two dates, Date 0 and Date 1. The risk-free rate between Date 0 and Date 1 is r. Consider a forward on a non-dividend-paying stock S.

1. Here we consider the payoff diagrams of the securities. All payoff diagrams should be expressed as a function of the underlying stock S.

   a) Draw the payoff diagram of a portfolio of K bonds.

   b) Draw the payoff diagram of the stock.

   c) Draw the payoff diagram of the forward.

2. Here we replicate the forward and find the forward price F.

   a) Use the payoff diagrams above to show graphically how to replicate the forward as a portfolio of bonds and stocks. (You will probably want to draw a few graphs on different axes and add them.)

   b) Use your answer above and the Law of One Price to write down a formula for the forward price.

3. Here we investigate comparative statics on the forward price.

   a) Suppose that the volatility of S increases, but that *all else is equal* (i.e. S_0 and r are unaffected). How do you expect the forward price to change? Is the assumption that "all else is equal" reasonable? How might it bias your answer? (Be brief.)

   b) Suppose that S becomes costly to store, but that *all else is equal* (i.e. S_0 and r are unaffected). How do you expect the forward price to change? Is the assumption that "all else is equal" reasonable? How might it bias your answer? (Be brief.)

IV. SWAPS

There are two dates, Date 0 and Date 1. There are two assets, Asset 1 and Asset 2. Let S_t^n denote the price of Asset n at Date t. There are *no* storage costs, dividends, etc.

Here, we study a swap, which is similar to a forward, but rather than agreeing to the cash purchase of an asset in the future, you agree to exchange another asset for it. Specifically, here, as in a forward, you exchange nothing today and you get one unit of S^1 at Date 1. Except instead of paying in cash, you deliver f units of Asset 2.

Here we find the "swap rate" f.

1. Keeping the Date-1 price of Asset 2 S_1^2 and the swap rate f fixed, draw the payoff diagram of the long position in the swap as a function of S_1^1.

2. Write down the Date-1 payoff of the long position in the swap in terms of S_1^1, S_1^2, and f.

3. Write down a replicating portfolio for the long swap.

4. Find an expression for the swap rate f. What result have you used?

5. Suppose that Asset 2 becomes riskier. Do you think the swap rate f increase, decreases, or stays the same? Explain briefly but carefully the assumptions behind and/or the limitations of your answer.

V. The Forward Curve and Forecasting

This question helps you understand terminology and common errors when working with the forward curve.

1. Read Section I.A of Gilje, Loutskina, and Murphy's 2019 paper "Drilling and Debt" available at https://www.ssrn.com/abstract=2939603

    a) Look at the forward curves in Figure 2. Suppose you have one barrel of oil. You can either sell it in the spot market today or sell it six months forward.

    In September 2014, how much more money do you (nominally) make if you sell it forward than if you sell it today (in percent)? In February 2015?

    b) What do the authors argue is the right strategy for oil firms?

    c) How do they say this strategy is affected by the forward curve? Why?

d) What other factors could affect the forward curve? Do they affect the authors' interpretation?

Hint: Cf. footnote 9.

2. Read Kaminska's 2011 blog post "All Together Now: The Forward Curve Is not a Forecast" available at https://ftalphaville.ft.com/2011/02/28/500181/all-together-now-the-forward-curve-is-not-a-forecast/

a) What is the common error that she describes?

b) What are the reasons she says it is an error? (Be brief.)

c) What would she say about Gilje, Loutskina, and Murphy's argument?

d) Who do you think is right?

VI. MGRM

1. Why did MGRM offer a price guarantee on purchase contracts for its customers?

2. What position did MGRM have in oil before its hedge (in rough terms)?

3. What was its hedging strategy? I.e. what did it use to hedge (in rough terms)?

4. Here we study MGRM's strategy quantitatively (you will probably want to use a program like Excel). Consider the forward curve on October 19, 1993. Suppose that they sold forward at a constant price, $F_0(T) \equiv 19$.

   a) Now suppose the price of oil stays constant at 20 (see Exhibit 1) over the next ten years. What is your monthly profit/loss from selling oil forward (the unhedged position), assuming you sold one barrel forward each month?

   b) Suppose that the interest rate is 7% (and the yield curve is flat). Calculate the present value of MGRM's short position, maintaining the assumption that the spot price stays constant. I.e. find the present value of the monthly P&L you found above. For this exercise, you should assume continuous compounding.

   c) Suppose that the spot price decreases by one dollar forever. How much does the value of your position change by?

5. With your forward position, you sold one barrel forward each month, for a total of 120 barrels. Suppose you hedge these all with a one-month futures contract, as MGRM did. I.e. you go long 120 one-month futures.

   a) Suppose the price of oil stays constant. What are your expected profits/losses from this position (at the end of the month)?

   b) As above, calculate the PV of this position.

      Hint: You have discount for only one month. (Why?)

   c) Suppose the spot price decreases by \$1. How much does the value of your position change by?

6. What happened to the value of MGRM's position in 1993? Why?

   Hint: Compare your answers for the change in the value of the initial position and the change in the value of the hedge.

7. If MGRM can endure these losses, is its strategy sustainable in the long run? Would you say that MGRM was just unlucky or was there something fundamentally wrong with MGRM's strategy? If there was something fundamentally wrong, what was it?

   Hint: Contrast your answers above to identify one key mismatch between MGRM's assets and their hedge.